

Volume 11 – Quantitative Research & Trading Methodology

AI Quant Trading Platform

Version: 1.0

Document Type: Quantitative Research & Trading Methodology

Classification: Trading Intelligence Foundation

1. Purpose

This document defines the quantitative research framework used to evaluate markets, validate strategies, optimize portfolio allocation, and measure long-term trading performance.

The purpose is to ensure that every trading strategy is based on measurable statistical evidence rather than intuition.

The platform should continuously evaluate opportunities using probability, statistics, and risk management before allowing any automated execution.

2. Research Principles

The platform follows these principles:

- Capital preservation is the first priority.
 - Risk-adjusted performance is more important than raw returns.
 - Every strategy must be statistically validated.
 - Decisions should be data-driven.
 - Models should be explainable and reproducible.
 - Historical success does not guarantee future performance.
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3. Market Theory

The platform should understand multiple market models.

Dow Theory

Used for:

- Primary Trend
 - Secondary Trend
 - Trend Confirmation
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Wyckoff Method

Recognize

- Accumulation
 - Distribution
 - Markup
 - Markdown
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Smart Money Concepts (SMC)

Analyze

- Liquidity
 - Order Blocks
 - Fair Value Gaps
 - Break of Structure
 - Change of Character
 - Premium & Discount Zones
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ICT Concepts

Support

- Liquidity Pools
 - Optimal Trade Entry
 - Session Timing
 - Institutional Order Flow
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Auction Market Theory

Evaluate

- Value Areas
- Acceptance
- Rejection
- Market Balance
- Imbalance

4. Market Regime Classification

The system continuously classifies markets into:

- Strong Bull Trend
- Weak Bull Trend
- Strong Bear Trend
- Weak Bear Trend
- Sideways Range
- High Volatility
- Low Volatility
- News-Driven Market

Strategies should activate only when compatible with the detected regime.

5. Probability Framework

Every opportunity receives probabilistic evaluation.

Measurements include:

- Confidence Score
- Expected Value
- Expected Risk
- Estimated Reward
- Historical Similarity
- Risk-Reward Ratio

The platform must avoid deterministic predictions and instead rank opportunities by relative quality.

6. Statistical Analysis

The research framework shall support:

Descriptive Statistics

- Mean
- Median
- Mode
- Standard Deviation
- Variance

Distribution Analysis

- Skewness

- Kurtosis
- Percentiles

Performance Statistics

- Win Rate
 - Loss Rate
 - Profit Factor
 - Average Win
 - Average Loss
 - Expectancy
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7. Risk Metrics

Portfolio risk evaluation includes:

- Maximum Drawdown
- Relative Drawdown
- Daily Drawdown
- Volatility
- Exposure
- Concentration Risk
- Correlation Risk

Risk metrics must be visible before trade execution.

8. Portfolio Allocation

Supported allocation models:

- Equal Weight
- Fixed Percentage
- Volatility Adjusted
- Risk Parity
- Maximum Diversification
- Custom Allocation

The platform should prevent excessive concentration in correlated assets.

9. Correlation Analysis

The engine measures:

- BTC Correlation
- ETH Correlation

- Stablecoin Correlation
- Sector Correlation
- Portfolio Correlation
- Rolling Correlation

Highly correlated positions may trigger exposure warnings.

10. Position Sizing

Supported methodologies:

- Fixed Position Size
- Fixed Risk Percentage
- ATR-Based Position Size
- Volatility-Based Position Size
- Portfolio Percentage
- Kelly Criterion (Experimental)

Position sizing is determined after strategy validation and risk approval.

11. Volatility Analysis

Measure:

- Historical Volatility
- Realized Volatility
- ATR
- Bollinger Width
- Volatility Regime

Strategies may adapt parameters according to current volatility.

12. Liquidity Analysis

Evaluate:

- Daily Trading Volume
- Bid-Ask Spread
- Order Book Depth
- Slippage Risk
- Large Order Impact

Markets with insufficient liquidity should be filtered before execution.

13. Strategy Validation

Every strategy must be evaluated using:

- Historical Backtesting
- Walk-Forward Testing
- Out-of-Sample Testing
- Paper Trading
- Forward Performance Monitoring

Production deployment requires successful validation.

14. Performance Metrics

Track:

- Sharpe Ratio
- Sortino Ratio
- Calmar Ratio
- Profit Factor
- Recovery Factor
- Maximum Drawdown
- Average Holding Time
- Average Trade Duration

Metrics should be available by strategy and portfolio.

15. Optimization Policy

Optimization should focus on robustness rather than maximum historical return.

Requirements:

- Avoid overfitting
 - Validate across multiple market conditions
 - Compare against baseline strategies
 - Record optimization history
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16. Research Workflow

Research Idea

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Historical Data Collection



Feature Engineering



Strategy Design



Backtesting



Walk-Forward Validation



Paper Trading



Risk Review



Production Approval

17. Decision Framework

Trade decisions must satisfy:

- Statistical edge
- Acceptable risk
- Strategy compatibility
- Portfolio compatibility
- Market regime compatibility
- Liquidity requirements

Failure in any critical category results in trade rejection.

18. Continuous Evaluation

The platform periodically reviews:

- Strategy performance
- Market regime changes
- Risk profile effectiveness
- Model drift
- Portfolio diversification

Recommendations for improvement are generated but require user or administrator approval before affecting production.

19. Research Governance

All research artifacts shall include:

- Version
- Author
- Dataset Version
- Validation Results
- Approval Status
- Deployment Date
- Rollback Reference

This ensures traceability and reproducibility.

20. Design Principles

- Prefer robust strategies over optimized historical performance.
- Measure risk before expected return.
- Validate every assumption with historical evidence.
- Use probability to prioritize opportunities, not to guarantee outcomes.
- Keep research reproducible and fully auditable.
- Separate experimental research from production execution.